

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0300

Muscoda Industrial 138 kV
June 23, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0300. The Interconnection Customer proposes a 294.7 MW solar photovoltaic facility interconnecting at the Muscoda Industrial 138 kV substation in the Virginia area. The request was received on December 31, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 185 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,158,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Muscoda Industrial - Elmont 138 kV is overloaded to 138.0% of its 200 MVA emergency rating for the loss of the Muscoda Industrial - Bear Island 138 kV L. The project contributes 100.0% DFAX (294.7 MW) to this overload.

Muscoda Industrial - Bear Island 138 kV L: 127.9% loading (216 MVA) following the outage of the Chancellor Row - Bear Island 345 kV; DFAX = 71.0%.

For the loss of Muscoda Industrial - Bear Island 138 kV, the Kimberley - Elmont Substation 138 kV loads to 123.8% of its 192 MVA rating. Project DFAX: 100.0%.

For the loss of Bear Island - Muscoda Industrial 138 kV, the Kimberley - Shack Mills Station 138 kV loads to 101.4% of its 213 MVA rating. Project DFAX: 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muscoda Industrial - Elmont 138 kV	Rebuild line with higher-capacity conductor	\$4,951,000	100% (\$4,951,000)
Muscoda Industrial - Bear Island Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,276,000	84% (\$3,603,000)
Kimberley - Elmont Substation 138 kV	Rebuild line with higher-capacity conductor	\$2,343,000	100% (\$2,343,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kimberley - Shack Mills Station 138 kV	Reconductor line	\$1,261,000	100% (\$1,261,000)
	Total allocated to project		\$12,158,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.